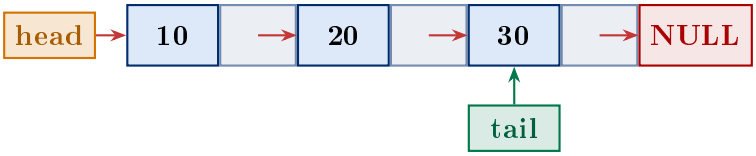
Linked Lists

From Absolute Basics to Advanced Mastery

Complete Study Notes with Your Own C++ Code



Files Covered in This Document

- `singlyLinkedList.cpp`
- `Assignment3Ques1.cpp`
- `Assignment3Ques2.cpp`
- `Assignment3Ques3.cpp`
- `Assignment3Ques4.cpp`
- `DoublyLinkedList.cpp`

github.com/SouvikGhorui/Data-Structure-and-Algorithm-with-CPP

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1. What is a Linked List?

1.1 The Core Idea

A **Linked List** is a linear data structure made of **nodes**. Each node stores two things:

1. **data** – the actual integer (or any value) you want to store
2. **next** – a pointer to the next node in the chain

The last node's `next` is `NULL` – meaning "nothing comes after."

Think of a linked list like a treasure hunt – each clue (node) tells you where the next clue (node) is. The last clue says "the treasure hunt is over" (`NULL`).

1.2 Your Node Class

In every file in your repo, you define a `Node` class like this:

```cpp
class Node {
public:
    int data;     // stores the value
    Node* next;   // stores address of the next node

    Node(int val) {
        data = val;
        next = NULL;    // you use NULL (C-style), which is fine in C++
    }
};
```

Listing 1.1: Node class – your style

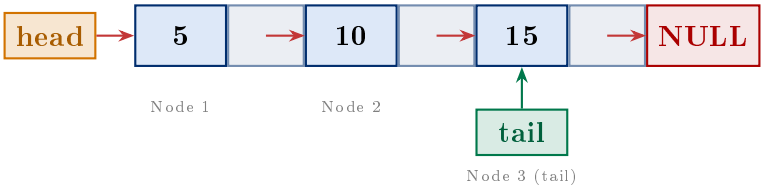
Note

You use `NULL` throughout your code. This is perfectly valid C++. Modern C++ (C++11 onwards) introduces `nullptr` which is type-safe, but `NULL` works correctly

for pointer comparisons.

1.3 Your LinkedList Class Structure

All your linked list classes carry both a `head` and a `tail` pointer. This is a smart design – it lets `push_back` run in $O(1)$ instead of $O(n)$!




```cpp
class LinkedList {
public:
    Node* head;
    Node* tail;

    LinkedList() {
        head = tail = NULL;  // empty list: both point to nothing
    }
    // ... methods below
};
```

Listing 1.2: LinkedList class with head and tail

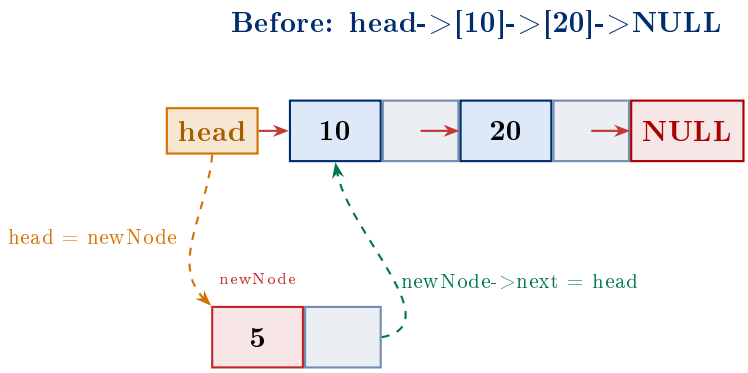
1.4 Why Two Pointers (head AND tail)?

Operation	Only head	head + tail
`push_front`	$O(1)$	$O(1)$
`push_back`	$O(n)$ – must walk to end	$O(1)$ – tail points there
`pop_back`	$O(n)$	$O(n)$ – still need to find second-to-last
`show`	$O(n)$	$O(n)$

2. Core File: singlyLinkedList.cpp

This is your foundational file. It implements all basic operations on a singly linked list. Let's go through every method.

2.1 push_front – Insert at Beginning

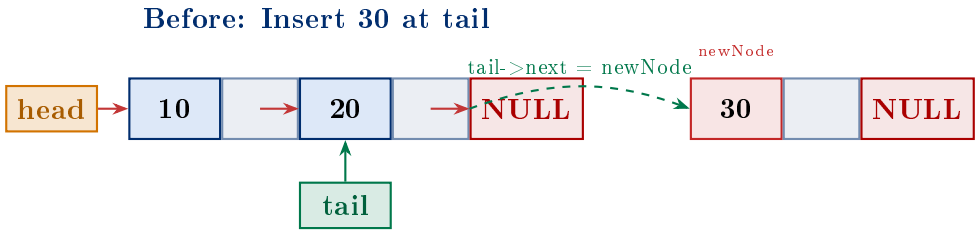



```cpp
void push_front(int val) {  // O(1)
    Node* newNode = new Node(val);
    if (head == NULL) {
        head = tail = newNode;  // first node: both head and tail point to it
        return;
    }
    else {
        newNode->next = head;   // new node points to old head
        head = newNode;         // head is updated to new node
    }
}
```

Listing 2.1: push_front – O(1)

Time: $O(1)$ – no traversal needed, just update head.
Space: $O(1)$ – only one new node allocated.

2.2 push_back – Insert at End




```cpp
void push_back(int val) {  // O(1)
    Node* newNode = new Node(val);
    if (head == NULL) {
        head = tail = newNode;  // empty list
    }
    else {
        tail->next = newNode;   // old tail links to new node
        tail = newNode;         // update tail to new node
    }
}
```

Listing 2.2: push_back – O(1) thanks to tail pointer

Because you store a `tail` pointer, `push_back` is $O(1)$. Without `tail`, you would walk the entire list each time – $O(n)$.

2.3 pop_front – Remove from Beginning

```cpp
void pop_front() {  // O(1)
    if (head == NULL) {
        cout << "There is nothing to delete!!";
        return;
    }
    Node* temp = head;        // save old head
    head = head->next;        // move head to next node
    temp->next = NULL;        // disconnect old head
    delete temp;              // free memory
}
```

Listing 2.3: pop_front – O(1)

2.4 pop_back – Remove from End

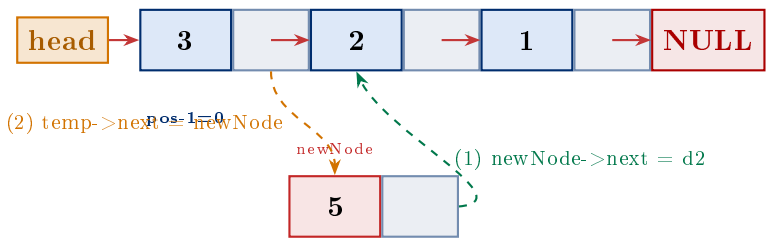
```cpp
void pop_back() {  // O(n)
    if (head == NULL) {
        cout << "There is nothing to delete!!";
        return;
    }
    Node* temp = head;
    while (temp->next != tail) {   // walk until node BEFORE tail
        temp = temp->next;
    }
    temp->next = NULL;   // second-to-last becomes new tail
    delete tail;
    tail = temp;
}
```

Listing 2.4: pop_back – O(n) for singly linked list

Why is pop_back O(n) even with a tail pointer? We know where the `tail` is, but we need to find the node *before* it to update its `next` to `NULL`. In a singly linked list, there is no `prev` pointer, so we must walk from `head`. This is why doubly linked lists (your `DoublyLinkedList.cpp`) solve this.

2.5 insert – Insert at Any Position

insert(5, pos=1): [3]->[2]->[1] – insert 5 after index 0



Source: `singlyLinkedList.cpp`

```cpp
void insert(int val, int pos) {  // O(n)
    if (pos < 0) {
        return;
    }
    else if (pos == 0) {
        push_front(val);          // reuse push_front for position 0
    }
    else {
```

```
        Node* newNode = new Node(val);
        Node* temp = head;
        for (int i = 0; i < pos - 1; i++) {   // walk to pos -1
            if (temp == NULL) {
                cout << "Invalid pos";
                return;
            }
            temp = temp->next;
        }
        newNode->next = temp->next;   // (1) link new node forward
        temp->next = newNode;         // (2) link prev node to new
    }
}
```

Listing 2.5: insert at position – O(n)

2.6 show – Display the List

```
void show() {  // O(n)
    Node* temp;
    temp = head;
    while (temp != NULL) {
        cout << temp->data << " -> ";   // print each node
        temp = temp->next;
    }
    cout << "NULL" << endl;  // marks end of list
}
// Output: 3 -> 5 -> 2 -> 1 -> NULL
```

Listing 2.6: show – O(n)

2.7 search – Find a Value

Source: singlyLinkedList.cpp

```
void search(int val) {  // O(n)
    Node* temp = head;
    while (temp != NULL) {
        int count = 0;                  // BUG: count is reset every iteration!
        if (temp->data == val) {
            cout << "Value is in the " << count << " th
```

```
    place" << endl;
            return;
        }
        temp = temp->next;
        count++;
    }
    cout << "Value is not in linkedlist" << endl;
}
```

Listing 2.7: search – O(n)

Bug in `search()` – `count` is declared inside the loop!
`int count = 0;` is inside the `while` loop, so it resets to `0` on every iteration. It will always print "position 0" even if the value is at position 5.
Fixed version:

```
void search(int val) {
    Node* temp = head;
    int count = 0;          // declare OUTSIDE the loop
    while (temp != NULL) {
        if (temp->data == val) {
            cout << "Value found at position " << count <<
    endl;
            return;
        }
        temp = temp->next;
        count++;            // only increment, never reset
    }
    cout << "Value not in linkedlist" << endl;
}
```

3. Assignment3Ques1.cpp – Search & Length

This file adds two important utility methods: `search` (returns `bool`) and `length`.

3.1 search – Boolean Version

This version is cleaner – it takes the `head` pointer as a parameter and returns `true`/`false`.

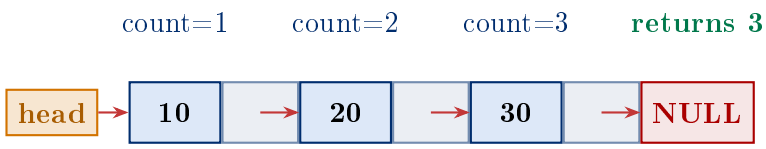
```cpp
bool search(Node* head, int x) {
    Node* temp = head;
    while (temp != NULL) {
        if (temp->data == x) {
            return true;     // found!
        }
        temp = temp->next;
    }
    return false;            // not found
}
```

Listing 3.1: search – boolean return, O(n)

This version is better than the one in `singlyLinkedList.cpp` because:

- Returns `bool` so the caller can use it in `if` statements
- No bug with `count`
- `main()` shows correct usage: `if(LL.search(LL.head, 4)) { ... }`

3.2 length – Count Nodes



```cpp
int length(Node* head) {
    Node* temp = head;
    int count = 0;
    while (temp != NULL) {
        count++;
        temp = temp->next;
    }
    return count;
}
```

Listing 3.2: length – count nodes, O(n)

3.3 Complete main() for Ques1

```cpp
int main() {
    int n, temp;
    cout << "Enter number of elements: ";
    cin >> n;
    LinkedList LL;
    for (int i = 0; i < n; i++) {
        cin >> temp;
        LL.push_back(temp);
    }
    if (LL.search(LL.head, 4)) {
        cout << "4 is in LinkedList" << endl;
    }
    LL.show();
    cout << "Total nodes = " << LL.length(LL.head) << endl;
    return 0;
}
```

Listing 3.3: main() – Ques1

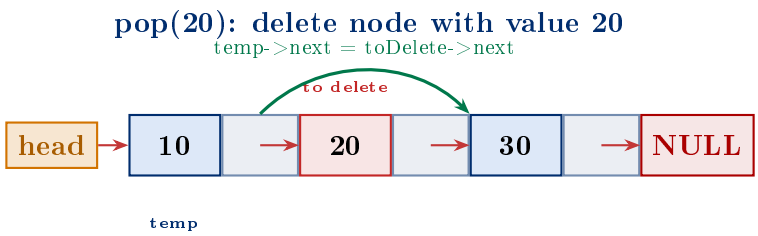
4. Assignment3Oues2.cpp – Deletion Methods

This file focuses on deletion: `pop_front`, `pop_back`, and `pop(x)` (delete by value).

4.1 pop_front and pop_back

Same as in `singlyLinkedList.cpp`. Covered in Chapter 2.

4.2 pop(x) – Delete Node by Value



Source: `Assignment3Oues2.cpp`

```cpp
void pop(int x) {
    if (head == NULL) {
        return;
    }
    // Special case: value is at head
    if (head->data == x) {
        pop_front();
        return;
    }
    Node* temp = head;
    // Walk until temp->next has the value x
    while ((temp->next->data != x) && (temp->next != NULL)) {
        temp = temp->next;
    }
    if (temp->next == NULL) {
        return;   // value not found
    }
    // Special case: value is at tail
    if (temp->next == tail) {
```

```
        pop_back();
        return;
    }
    // General case: unlink and delete
    Node* toDelete = temp->next;
    temp->next = toDelete->next;
    delete toDelete;
    return;
}
```

Listing 4.1: pop(x) – delete node by value, O(n)

Potential Crash in the `while` condition!
The condition is:

```
while ((temp->next->data != x) && (temp->next != NULL))
```

If `temp->next` is `NULL` and we evaluate `temp->next->data`, the program will crash with a **segmentation fault** (null pointer dereference). C++ evaluates left-to-right, so `temp->next->data` runs before the `NULL` check.
Fixed version – swap the condition order:

```
while (temp->next != NULL && temp->next->data != x)
//      ^^^ check NULL first ^^^
```

4.3 Complete main() for Oues2

```
int main() {
    int n, temp;
    cout << "Enter number of elements: ";
    cin >> n;
    LinkedList LL;
    for (int i = 0; i < n; i++) {
        cin >> temp;
        LL.push_back(temp);
    }
    LL.show();
    LL.pop(3);      // delete node containing value 3
    LL.show();
    return 0;
}
```

Listing 4.2: main() – Oues2

5. Assignment3Ques3.cpp – Reverse & Middle

This file implements two classic interview problems: printing the list in reverse (recursively) and finding the middle element.

5.1 ReverseLL – Recursive Reverse Print

```cpp
void ReverseLL(Node* head) {
    if (head == NULL) {
        return;            // base case: end of list
    }
    ReverseLL(head->next);  // recurse FIRST (go to the end)
    cout << head->data << " ";  // print on the WAY BACK
}
```

Listing 5.1: ReverseLL – recursive reverse print, O(n)

How recursion prints in reverse – call stack visualised:

List: 1 -> 2 -> 3 -> NULL

Call 1: ReverseLL(1) – calls ReverseLL(2) first

Call 2: ReverseLL(2) – calls ReverseLL(3) first

Call 3: ReverseLL(3) – calls ReverseLL(NULL)

Call 4: ReverseLL(NULL) – base case, RETURN

Unwinding: prints 3, then 2, then 1 => output: 3 2 1

`ReverseLL()` **does not actually reverse the list**. It only *prints* values in reverse order. The original list stays unchanged. To actually reverse the list (modify pointers), you need the three-pointer iterative approach shown in Chapter 8.

Time: $O(n)$ – every node visited once.
Space: $O(n)$ – recursion uses the call stack. One stack frame per node.

5.2 middleEle – Find the Middle

Your approach: compute the full length, then walk to `length/2`.

List: 10->20->30->40->50, length=5, length/2=2, middle=30

```cpp
int lengthLL(Node* head) {
    Node* temp = head;
    if (temp == NULL) return 0;
    int count = 0;
    while (temp != NULL) {
        count++;
        temp = temp->next;
    }
    return count;
}

void middleEle(Node* head) {
    Node* temp = head;
    for (int i = 0; i < lengthLL(temp) / 2; i++) {
        temp = temp->next;     // advance lengthLL/2 steps
    }
    cout << "Middle element = " << temp->data;
}
```

Listing 5.2: middleEle and lengthLL – O(n)

Complexity Analysis

Your approach: $O(n)$ for `lengthLL` + $O(n/2)$ for the walk = $O(n)$ total. Calls `lengthLL` once per call to `middleEle`.
Faster single-pass approach (Floyd's tortoise and hare):

```cpp
void middleEle_fast(Node* head) {
    Node* slow = head;
    Node* fast = head;
    while (fast != NULL && fast->next != NULL) {
        slow = slow->next;
```

```
            fast = fast->next->next;
        }
        cout << "Middle = " << slow->data;
    }
```

One pass, no extra function call – still $O(n)$ but runs faster in practice.

6. Assignment3Ques4.cpp – Remove & Re-insert

This file introduces two new ideas: `push_front`, `insert` (taking a `Node*`), and the interesting `remove_insert` function.

6.1 insert(pos, Node*) – Insert Using a Node Pointer

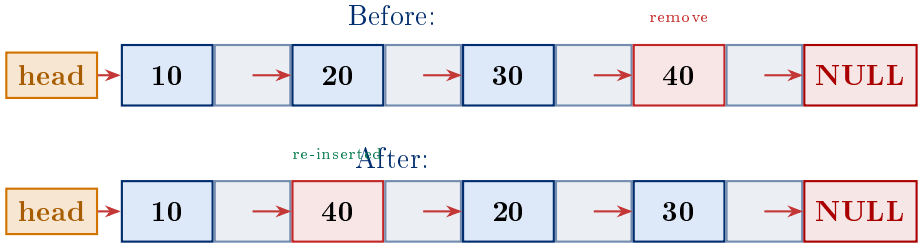
```
void insert(int pos, Node* node) {
    int count = 0;
    Node* temp = head;
    if (pos == 0) {
        push_front(node->data);
        return;
    }
    while (count != pos - 1) {      // walk to pos-1
        temp = temp->next;
        count++;
    }
    node->next = temp->next;        // link new node forward
    temp->next = node;              // link prev node to new
}
```

Listing 6.1: insert – takes position and Node pointer

6.2 remove_insert – Move a Node from One Position to Another

This is a unique function you wrote. It removes a node from `remove_pos` and re-inserts it at `insert_pos`.

remove_insert(3, 1): remove index 3, insert at index 1




```cpp
void remove_insert(int remove_pos, int insert_pos) {
    Node* temp = head;
    int count = 0;

    // Step 1: walk to node BEFORE remove_pos
    if (remove_pos == 0) {
        head = head->next;
    }
    else {
        while (count != remove_pos - 1) {
            temp = temp->next;
            count++;
        }
    }

    // Step 2: save the node to be moved, then unlink it
    Node* tempNode = new Node(temp->next->data);  // copy the data
    Node* temp2    = temp->next;
    temp->next     = temp2->next;
    delete temp2;

    // Step 3: re-insert at the desired position
    insert(insert_pos, tempNode);
}
```

Listing 6.2: remove_insert – O(n)

How remove_insert works step by step:

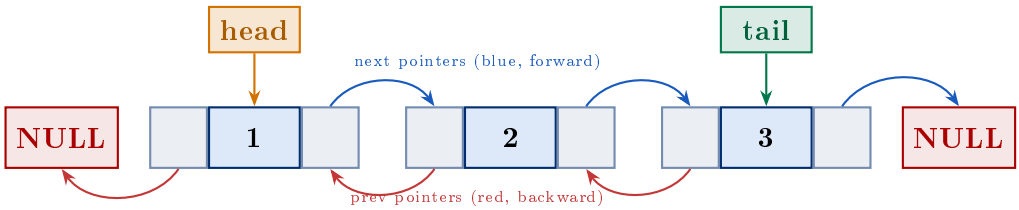
1. Walk to the node at `remove_pos - 1` (one before the target).
2. Copy the target node's data into a new node (`tempNode`).
3. Unlink the original node from the list and `delete` it.
4. Call `insert(insert_pos, tempNode)` to place the copy at the new position.

7. DoublyLinkedList.cpp – Doubly Linked List

7.1 What is a Doubly Linked List?

Your `DoublyLinkedList.cpp` introduces a `Node` with *three* fields: `prev`, `data`, and `next`. This lets you traverse both forwards and backwards.

Your DLL: each node has prev and next pointers



7.2 Your Node Class for DLL

```cpp
class Node {
public:
    int data;
    Node* next;
    Node* prev;          // NEW: backward pointer

    Node(int val) {
        data = val;
        next = NULL;
        prev = NULL;
    }
};
```

Listing 7.1: DLL Node with prev and next

7.3 push_front – DLL version

```cpp
void push_front(int val) {
    Node* newNode = new Node(val);
    if (head == NULL) {
        head = tail = newNode;
    }
    else {
        newNode->next = head;     // new node's next = old head
        head->prev    = newNode; // old head's prev = new node
        head          = newNode; // update head
    }
}
```

Listing 7.2: DLL push_front – O(1)

7.4 push_back – DLL version

```cpp
void push_back(int val) {
    Node* newNode = new Node(val);
    if (head == NULL) {
        head = tail = newNode;
    }
    else {
        newNode->prev = tail;     // new node's prev = old tail
        tail->next    = newNode; // old tail's next = new node
        tail          = newNode; // update tail
    }
}
```

Listing 7.3: DLL push_back – O(1)

7.5 pop_front – DLL version

Source: DoublyLinkedList.cpp

```cpp
void pop_front() {
    if (head == NULL) return;
    Node* temp = head;
    head = head->next;
```

```
    if (head != NULL) {
        head->prev = NULL;     // new head has no previous node
    }
    temp->next = NULL;
    delete temp;
}
```

Listing 7.4: DLL pop_front – O(1)

7.6 pop_back – DLL version: now truly O(1)!

```
void pop_back() {
    if (head == NULL) return;
    Node* temp = tail;
    tail = tail->prev;          // move tail back using prev pointer
    if (tail != NULL) {
        tail->next = NULL;      // new tail has no next
    }
    temp->prev = NULL;
    delete temp;
}
```

Listing 7.5: DLL pop_back – O(1) thanks to prev pointer!

This is the key advantage of DLL! In `singlyLinkedList.cpp`, `pop_back` is $O(n)$ because you must walk to find the second-to-last node. In DLL, `tail->prev` gives you the second-to-last node directly in $O(1)$.

7.7 show – DLL version

Source: `DoublyLinkedList.cpp`

```
void show() {
    Node* temp = head;
    while (temp != NULL) {
        cout << temp->data << "<=>";    // your style: <=> separator
        temp = temp->next;
    }
    cout << "NULL" << endl;
}
```

```
// Output: 1<=>2<=>3<=>NULL
```

Listing 7.6: DLL show

7.8 Complete DLL main()

```
int main() {
    doublyList dll;
    dll.push_back(1);
    dll.push_back(2);
    dll.push_back(3);
    dll.pop_back();        // removes 3
    dll.show();            // Output: 1<=>2<=>NULL
    return 0;
}
```

Listing 7.7: main() – DoublyLinkedList.cpp

8. SLL vs DLL – Comparison from Your Code

8.1 Operations Compared

Operation	Your SLL	Your DLL	Why different?
`push_front`	$O(1)$	$O(1)$	Same logic
`push_back`	$O(1)$	$O(1)$	Both use `tail` pointer
`pop_front`	$O(1)$	$O(1)$	Same logic
`pop_back`	$O(n)$	$O(1)$	DLL has `prev` pointer
`insert`	$O(n)$	$O(n)$	Must search
`search`	$O(n)$	$O(n)$	Must traverse
`show`	$O(n)$	$O(n)$	Full traversal
Memory per node	1 ptr	2 ptrs	DLL uses more RAM

8.2 Structural Comparison

Property	SLL (your files)	DLL (your file)
Class name	`LinkedList` / `List`	`doublyList`
Node fields	`data, next`	`data, next, prev`
Separator in `show`	`" -> "`	`"<=>"`
Backward traversal	No	Yes
Delete without search	No	Yes (if you have the pointer)

9. Advanced Operations

9.1 Actual List Reversal – Three Pointer Technique

Your `ReverseLL` in Ques3 only *prints* in reverse. This version actually *modifies* the list using your coding style:

```
void reverseList() {
    Node* prev = NULL;
    Node* curr = head;
    Node* next = NULL;
    tail = head;            // old head becomes new tail

    while (curr != NULL) {
        next        = curr->next;   // save next
        curr->next  = prev;         // flip pointer
        prev        = curr;         // advance prev
        curr        = next;         // advance curr
    }
    head = prev;            // old tail becomes new head
}
```

Listing 9.1: True reverse – modifies the list

Dry run on your list [1 → 2 → 3 → NULL]:

Step	prev	curr	next	What happens
Start	NULL	1	–	initialise
1	NULL	1	2	1->NULL; prev=1; curr=2
2	1	2	3	2->1; prev=2; curr=3
3	2	3	NULL	3->2; prev=3; curr=NULL
End	3	NULL	–	head = 3

9.2 Floyd's Cycle Detection

```
bool hasCycle() {
    Node* slow = head;
    Node* fast = head;

    while (fast != NULL && fast->next != NULL) {
```

```
        slow = slow->next;          // 1 step
        fast = fast->next->next;    // 2 steps

        if (slow == fast) return true;  // met inside cycle
    }
    return false;
}
```

Listing 9.2: Cycle detection – two pointer technique

9.3 Fast Middle Element (No Length Needed)

Improvement over your `middleEle` in Ques3:

```
void middleEle_fast(Node* head) {
    Node* slow = head;
    Node* fast = head;

    while (fast != NULL && fast->next != NULL) {
        slow = slow->next;          // 1 step
        fast = fast->next->next;    // 2 steps
    }
    // when fast hits end, slow is at middle
    cout << "Middle element = " << slow->data;
}
```

Listing 9.3: Fast middle – single pass, no lengthLL call

Your version: $O(n)$ for `lengthLL` + $O(n/2)$ walk = **1.5n** steps.
Fast version: Single pass, exactly **n/2** steps. Same $O(n)$ class but twice as fast in practice.

9.4 Merge Sort on Linked List

```
Node* getMiddle(Node* head) {
    Node* slow = head;
    Node* fast = head->next;
    while (fast != NULL && fast->next != NULL) {
        slow = slow->next;
        fast = fast->next->next;
    }
    return slow;    // slow is at mid
}

Node* merge(Node* a, Node* b) {
    if (a == NULL) return b;
    if (b == NULL) return a;
    if (a->data <= b->data) {
```

```
        a->next = merge(a->next, b);
        return a;
    }
    b->next = merge(a, b->next);
    return b;
}

Node* mergeSort(Node* head) {
    if (head == NULL || head->next == NULL)
        return head;
    Node* mid        = getMiddle(head);
    Node* secondHalf = mid->next;
    mid->next        = NULL;            // split
    return merge(mergeSort(head), mergeSort(secondHalf));
}
// Usage: LL.head = mergeSort(LL.head);
```

Listing 9.4: Merge Sort – O(n log n)

10. Bug Summary and Fixes

10.1 All Bugs Found in Your Files

File	Function	Bug	Fix
singlyLinkedList	`search()`	`int count = 0` inside loop, resets every iteration	Move `int count = 0` to before the `while` loop
Assignment3Oues2	`pop(x)`	`temp->next->data` evaluated before `NULL` check – crash risk	Swap condition order to check `temp->next != NULL` first
Assignment3Ques4	`remove_insert()`	When `remove_pos == 0` the code runs `head = head->next` but the original head is not deleted – memory leak	Save old head, then delete it
Assignment3Ques3	`ReverseLL()`	Comment in `main()` says "ReverseLL" but it is commented out – not called	Un-comment to use it, and note it only prints, does not reverse

10.2 Fixed pop() – Safe Version

```
void pop(int x) {
    if (head == NULL) return;

    if (head->data == x) {
        pop_front();
        return;
    }

    Node* temp = head;
    // FIX: check NULL before accessing ->data
    while (temp->next != NULL && temp->next->data != x) {
        temp = temp->next;
    }

```

```
    if (temp->next == NULL) return;    // not found

    if (temp->next == tail) {
        pop_back();
        return;
    }

    Node* toDelete = temp->next;
    temp->next = toDelete->next;
    delete toDelete;
}
```

Listing 10.1: pop(x) – fixed safe version

10.3 Fixed search() – Correct Position Counter

```
void search(int val) {
    Node* temp  = head;
    int   count = 0;            // FIX: declare OUTSIDE the while loop

    while (temp != NULL) {
        if (temp->data == val) {
            cout << "Value is in the " << count << " th place" << endl;
            return;
        }
        temp = temp->next;
        count++;
    }
    cout << "Value is not in linkedlist" << endl;
}
```

Listing 10.2: search() – fixed, count declared outside loop

11. Complete Complexity Reference

11.1 All Your Methods at a Glance

Method	File	Time	Space
`push_front`	SLL files	$O(1)$	$O(1)$
`push_back`	SLL files	$O(1)$	$O(1)$
`pop_front`	SLL files	$O(1)$	$O(1)$
`pop_back`	SLL files	$O(n)$	$O(1)$
`pop_back`	DoublyLinkedList	$O(1)$	$O(1)$
`insert`	SLL files	$O(n)$	$O(1)$
`pop(x)`	Oues2	$O(n)$	$O(1)$
`search`	Ques1, SLL	$O(n)$	$O(1)$
`length/lengthLL`	Ques1, Ques3	$O(n)$	$O(1)$
`ReverseLL`	Ques3	$O(n)$	$O(n)$
`middleEle`	Ques3	$O(n)$	$O(1)$
`remove_insert`	Ques4	$O(n)$	$O(1)$
`show`	All files	$O(n)$	$O(1)$